

Contents

0.1	Dataset Info	2
0.2	Analysis of Dataset	2
0.3	Observation	7
0.1	Objective	8
0.2	Classification of Dataset	8
0.1	Frequency Distribution Table	9
0.2	Representation	9
0.3	Observation	12
0.1	Calculating measure of variability	13
0.2	Representation	14
0.3	Observation	16
0.1	Calculating measure of variability	17
0.2	Representation	18
0.3	Observation	20

Experiment 1

Collect a real-life dataset and identify different types of data (numerical and categorical). Analyse the dataset and present it using suitable tables or charts.

1.1. Dataset Info

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import random
4
5 df_student = pd.read_csv("dataset.csv", delimiter=',',
6                           header='infer')
7 df_student.head()
```

Output:

	Name	Gender	Grade	StudyHours	Marks
0	Aarav	M	A	6	91
1	Ishita	F	A	7	95
2	Rohan	M	B	4	75
3	Meera	F	B	5	78
4	Kabir	M	C	2	55

`head()` function is used to print the first 5 records of the dataframe. We can use `dtypes` attribute to give out datatype of each column.

```
1 df_student.dtypes
```

Output:

```
Name          object
Gender         object
Grade          object
StudyHours     int64
Marks          int64
Attendance     int64
dtype: object
```

The dataset we chose contains 50 records. These records were Name, Gender, Grade, StudyHours, Marks & Attendance.

- **Numerical:** StudyHours, Marks, Attendance
- **Categorical:** Gender (M/F), Grade (A/B/C/D)

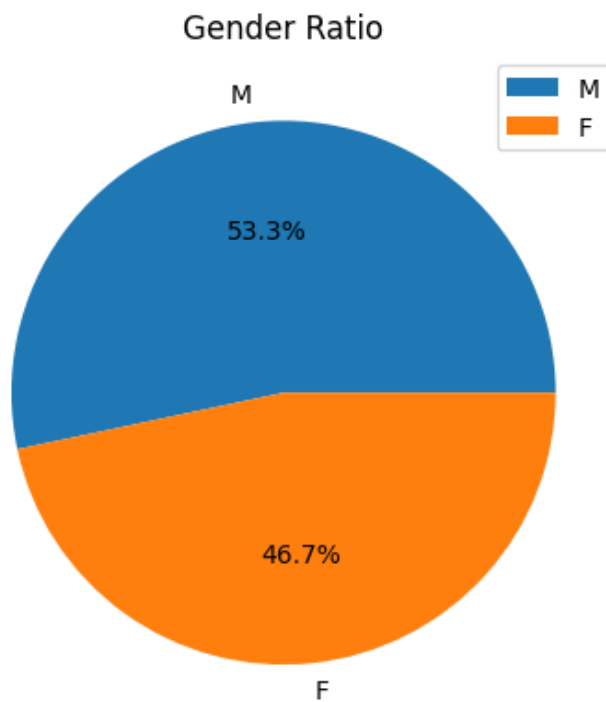
1.2. Analysis of Dataset

Let's do some basic analysis of the dataset using charts and graphs.

i) Gender Distribution

```
1 counts = df_student.Gender.value_counts()
2 plt.pie(counts, labels=counts.index, autopct='%1.1f%%')
3 plt.legend()
4 plt.title("Gender Ratio")
5 plt.show()
```

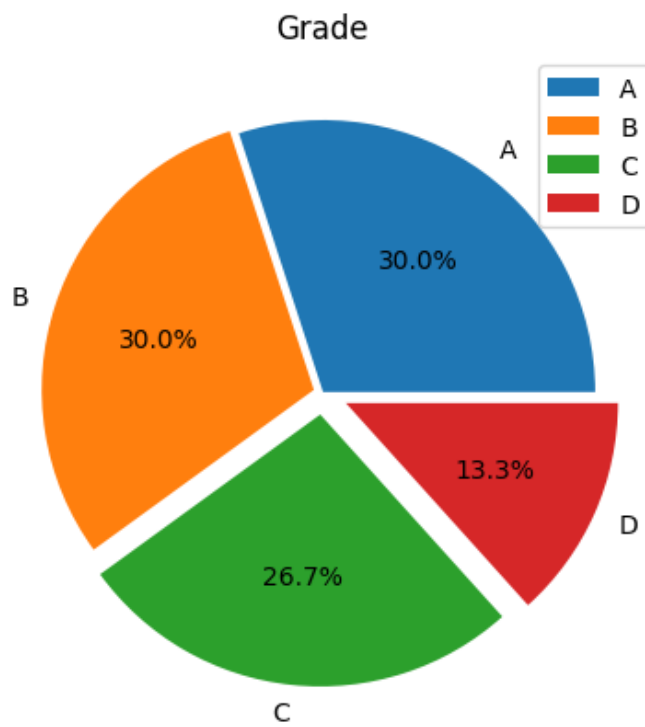
Output:



ii) Grade Distribution

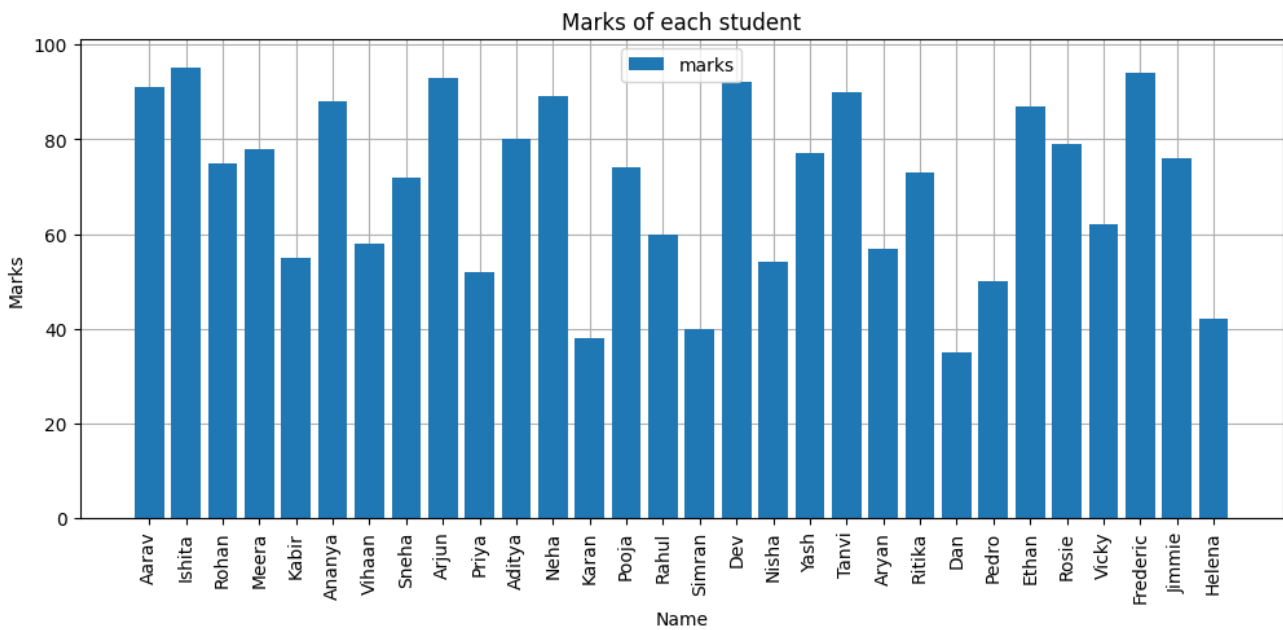
```
1 counts = df_student.Grade.value_counts()
2 plt.pie(counts, labels=counts.index, autopct='%1.1f%%',
3         explode=[random.uniform(0, 0.1) for _ in range(4)])
4 plt.legend()
5 plt.title("Grade")
6 plt.show()
```

Output:



iii) Marks of Each Student

```
1 plt.figure(figsize=(12,4.8))
2 plt.bar(df_student.Name, df_student.Marks, label="marks", zorder=2)
3
4 plt.title("Marks of each student")
5 plt.ylabel("Marks")
6 plt.xlabel("Name")
7 plt.ylim(top=101)
8 plt.xticks(rotation=90)
9 plt.grid(zorder=0)
10 plt.legend()
11
12 plt.show()
```

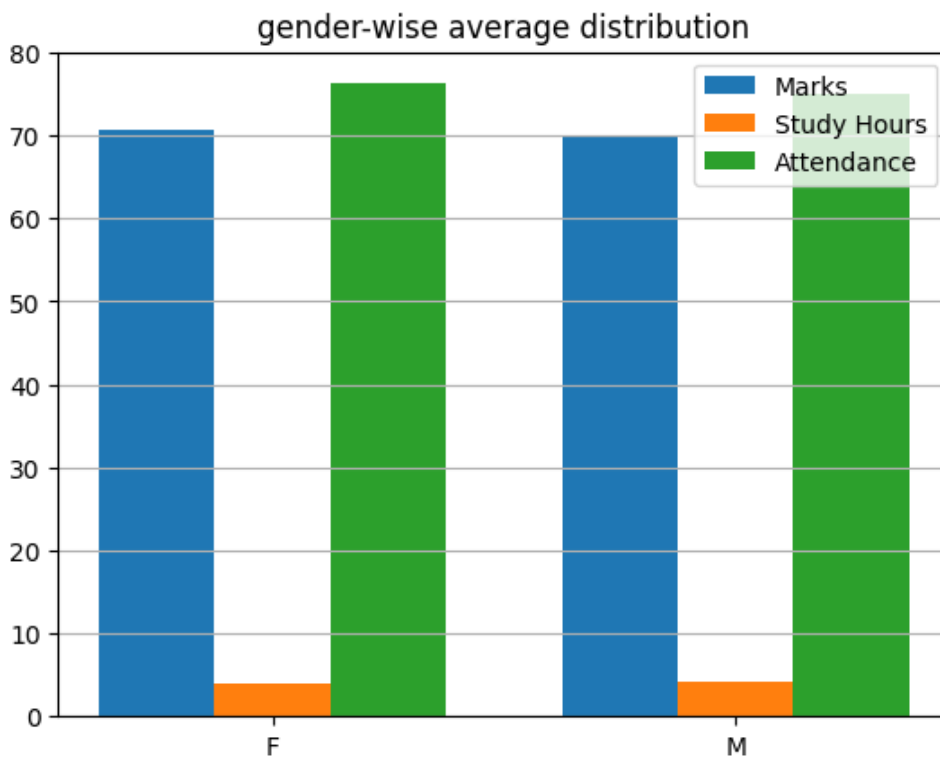


iv) Gender-wise average distribution

```

1 marks = df_student.groupby("Gender")["Marks"].mean()
2 studyhour = df_student.groupby("Gender")["StudyHours"].mean()
3 hardiksassignment =
4     df_student.groupby("Gender")['Attendance'].mean()
5
6 x = np.arange(marks.count())
7 width = 0.25
8
9 plt.bar(x-width, marks.values, width, label="Marks")
10 plt.bar(x, studyhour.values, width, label="Study Hours")
11 plt.bar(x + width, hardiksassignment.values, width,
12         label="Attendance")
13
14 plt.xticks(x, marks.index)
15 plt.grid(axis='y', zorder=0)
16 plt.legend()
17 plt.title("gender-wise average distribution")
18
19 plt.show()

```

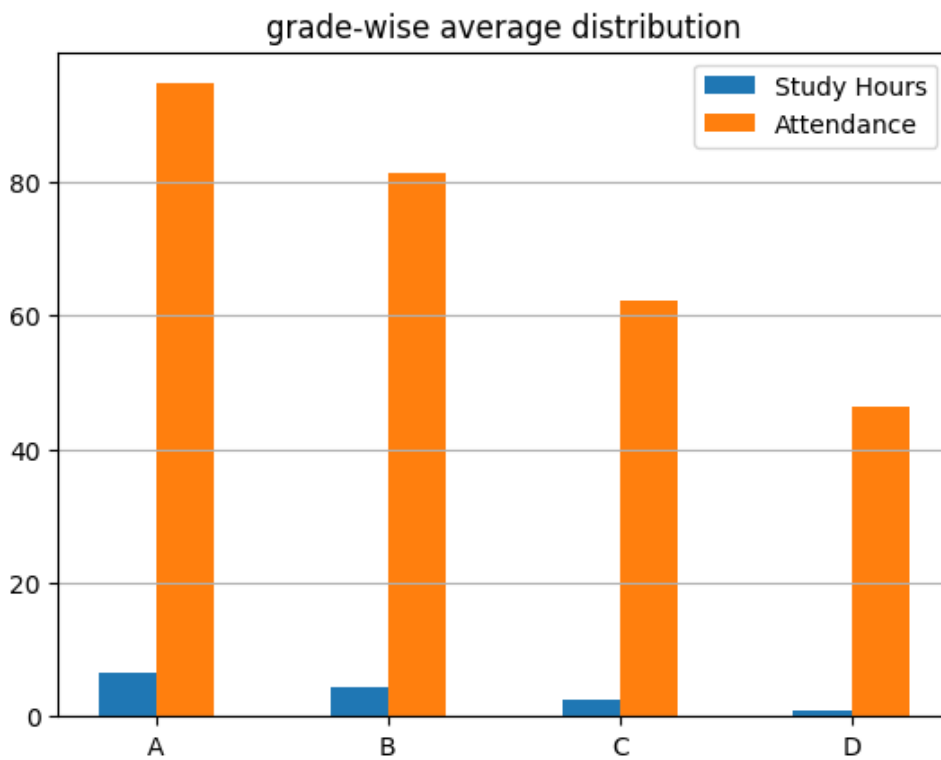


v) Grade-wise average distribution

```

1 studyhour = df_student.groupby("Grade")["StudyHours"].mean()
2 hardiksassignment = df_student.groupby("Grade")['Attendance'].mean()
3
4 x = np.arange(marks.count())
5 width = 0.25
6
7 plt.bar(x - width/2, studyhour.values, width, label="Study Hours")
8 plt.bar(x + width/2, hardiksassignment.values, width,
9         label="Attendance")
10
11 plt.xticks(x, marks.index)
12 plt.grid(axis='y', zorder=0)
13 plt.legend()
14 plt.title("grade-wise average distribution")
15 plt.show()

```



1.3. Observation

From these graphs and charts, we can deduce that this batch has an approximately equal ratio of males and females. Both groups have fairly similar average marks, study hour and attendance. Looking at the grade-wise distribution, we can see that as the grade decreases the study hours and attendance have also reduced.

Experiment 2

Take a dataset and classify the data based on measurement scales(nominal, ordinal, interval, and ratio). Provide appropriate examples for each category.

2.1. Objective

The objective of this experiment is to analyze and classify the dataset into four fundamental levels of measurement scale.

2.2. Classification of Dataset

	Variable	Data Type	Measurement Scale	Justification
1	Name	Categorical	Nominal	no mathematical relation or order
2	Gender	Categorical	Nominal	Two categories, no relation
3	Grade	Categorical	Ordinal	$A > B > C > D > E$, intervals are unequal
4	Marks	Numerical	Ratio	0 is nil, 90 marks is twice 45 marks
5	StudyHours	Numerical	Ratio	0 means nil
6	Attendance	Numerical	Ratio	0% means nil

Experiment 3

Construct a frequency distribution table from a given dataset by grouping the data into class intervals. Represent the data using graphs such as histograms or bar charts.

3.1. Frequency Distribution Table

We can group the dataset into class intervals by using 'cut()' function of pandas.

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 df_student = pd.read_csv("dataset.csv", delimiter=',',
6                           header='infer')
7
8 sturges = round(1+np.log2(df_student.Marks.count()))
9
10 counts, bins = np.histogram(df_student.Marks, bins=sturges)
11
12 freq_dis = pd.DataFrame({
13     'Class Interval': [f'{bins[i-1]} - {bins[i]}' for i in range(1,
14         len(bins))],
15     'Frequency': counts,
16     'Relative frequency': (counts / len(df_student)).round(2),
17     'Cumulative Frequency': counts.cumsum()
18 })
19
20 print(freq_dis)
```

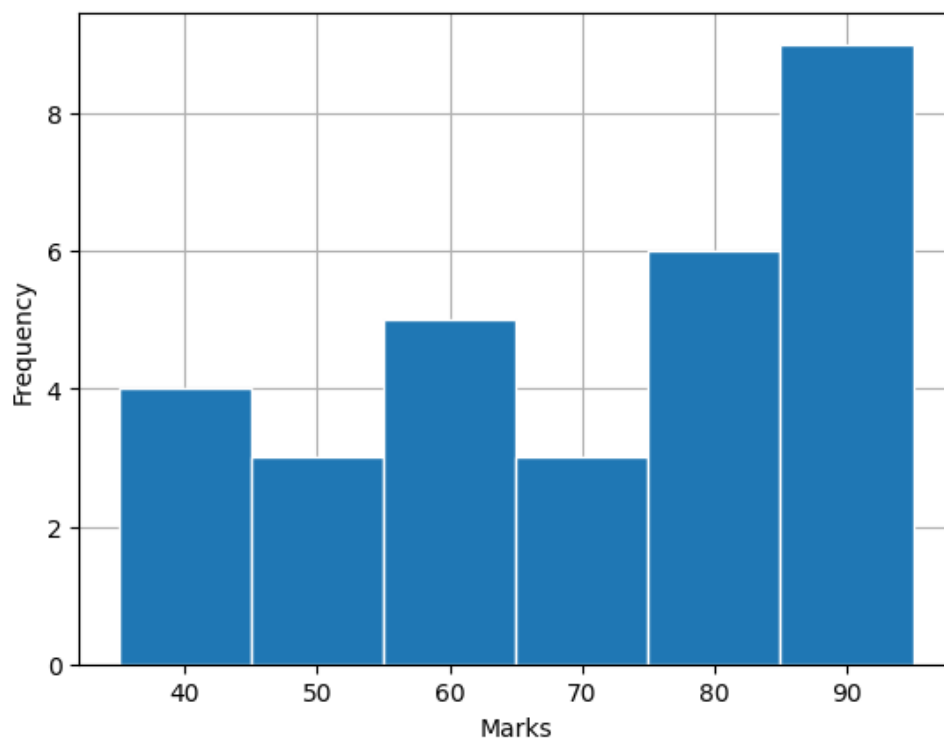
Output:

	Class Interval	Frequency	Relative Frequency	Cumulative Frequency
0	35.0 - 45.0	4	0.13	4
1	45.0 - 55.0	3	0.10	7
2	55.0 - 65.0	5	0.17	12
3	65.0 - 75.0	3	0.10	15
4	75.0 - 85.0	6	0.20	21
5	85.0 - 95.0	9	0.30	30

3.2. Representation

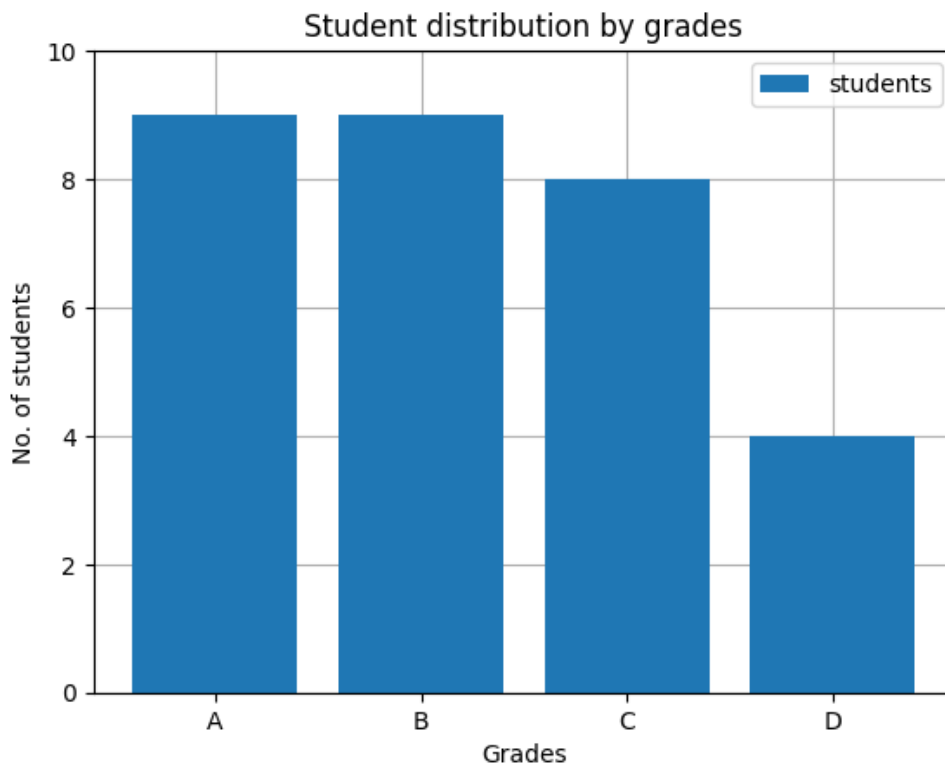
i) Histogram of marks records:

```
1 plt.hist(df_student.Marks, ec='white', bins=sturges, zorder=2)
2
3 plt.xlabel('Marks')
4 plt.ylabel('Frequency')
5 plt.grid(zorder=0)
6
7 plt.show()
```



ii) Grade distribution:

```
1 graded = df_student.groupby("Grade").Marks
2
3 plt.bar(graded.count().index, graded.count().values, label =
4         "students", zorder=2)
5
6 plt.title("Student distribution by grades")
7 plt.xlabel("Grades")
8 plt.ylabel("No. of students")
9 plt.ylim((0 ,10))
10 plt.legend()
11 plt.grid(zorder=0)
12 plt.show()
```

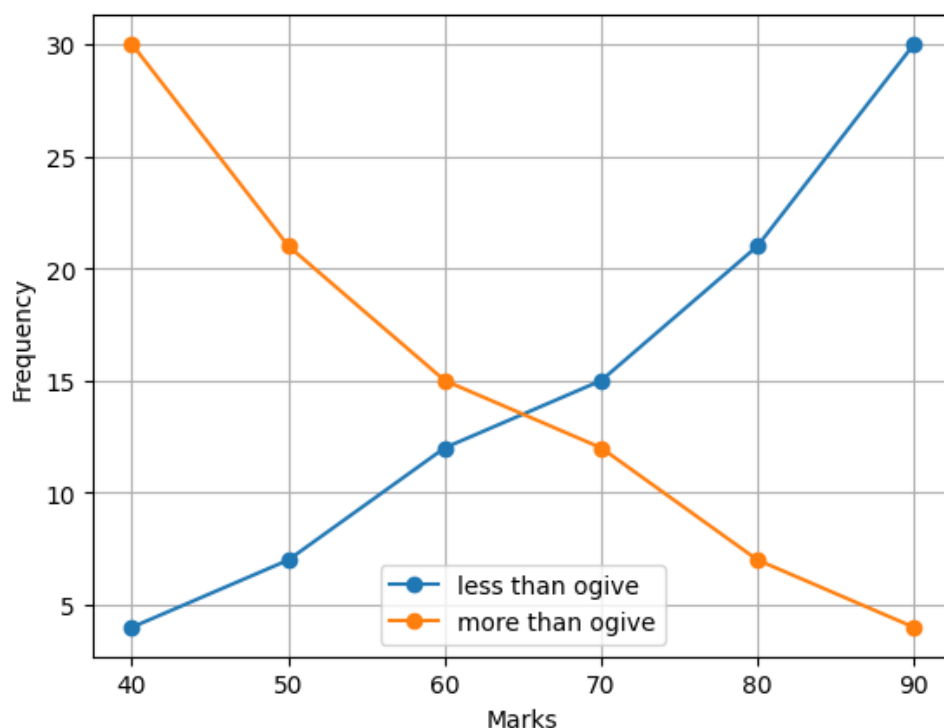


iii) Ogive graph

```

1 counts, bins = np.histogram(df_student.Marks, bins=sturges)
2
3 bin_centre = []
4 for i in range(1, len(bins)):
5     bin_centre.append((bins[i-1] + bins[i])/2)
6
7 plt.plot(bin_centre, counts.cumsum(), "o-", label="less than
8     ogive", zorder=2)
9
10 plt.plot(bin_centre, counts.cumsum()[::-1], "o-", label="more than
11     ogive", zorder=3)
12
13 plt.xlabel('Marks')
14 plt.ylabel('Frequency')
15 plt.legend()
16 plt.grid(zorder=0)
17 plt.show()

```



3.3. Observation

- i) From the frequency distribution table, the majority of students lies between 85 & 95 marks range, with the highest frequency of 9.
- ii) The histogram is slightly left skewed, indicating most students scored good marks except a few.
- iii) From the grade-wise distribution, grades A and B both have 9 students each, accounting for 60% of the class.
- iv) The ogive graph shows that 21/30 students scored below 85 marks.
- v) The combined relative frequency of 75.0 to 95.0 interval is 50%, meaning half of the class scored high marks. This indicates a strong overall performance.

Experiment 4

Calculate measure of variability such as range, variance and standard deviation for a dataset. Interpret what these measure indicate about the spread of the data.

4.1. Calculating measure of variability

i) Range

range = max - min

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 df_student = pd.read_csv("dataset.csv", delimiter=',',
6                           header='infer')
7
8 minval = df_student[['Marks', 'StudyHours', 'Attendance']].min()
9 maxval = df_student[['Marks', 'StudyHours', 'Attendance']].max()
10
11 rng = pd.DataFrame({
12     'Attribute': minval.index,
13     'Minimum': minval.values,
14     'Maximum': maxval.values,
15     'Range': (maxval - minval).values
16 })
17 print(rng)
```

Output:

	Attribute	Minimum	Maximum	Range
0	Marks	35	95	60
1	StudyHours	1	7	6
2	Attendance	42	99	57

ii) Variance

```
1 print(df_student[['Marks', 'StudyHours',
2                   'Attendance']].var().round(3))
```

Output:

Marks	345.752
StudyHours	3.995
Attendance	301.357

iii) Standard Deviation

```

1 df_student[['Marks', 'StudyHours', 'Attendance']].agg(['std',
    'mean']).T.round(3)

```

Output:

	std	mean
Marks	18.594	70.200
StudyHours	1.999	4.067
Attendance	17.360	75.567

4.2. Representation

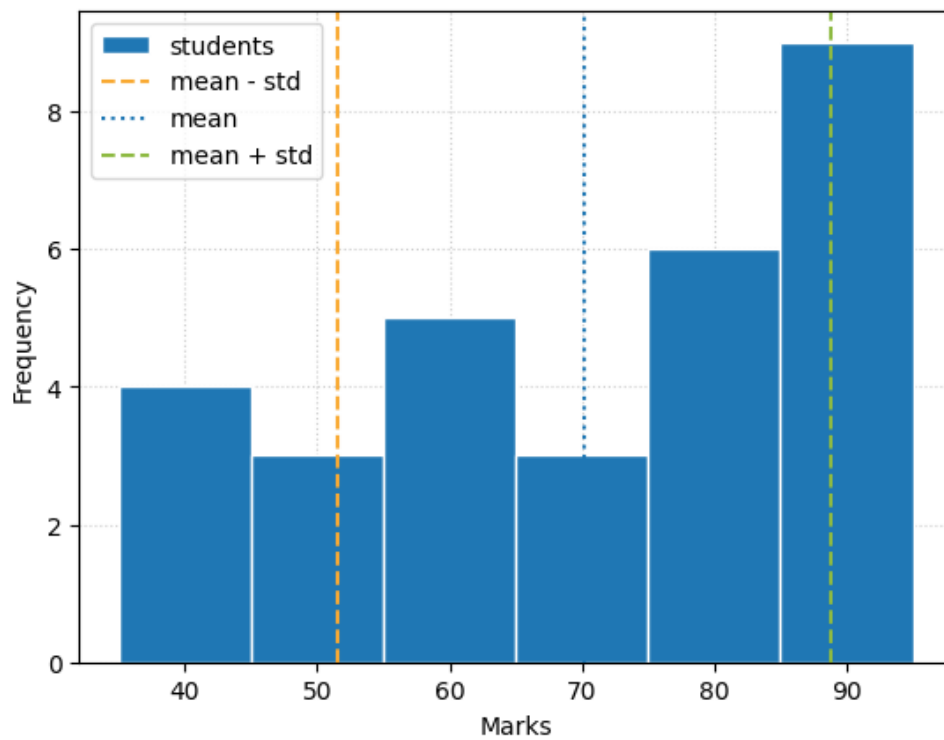
i) Spread of Marks

```

1 sturges = round(1+np.log2(df_student.Marks.count()))
2
3 mean = df_student.Marks.mean()
4 std = df_student.Marks.std()
5
6 plt.hist(df_student.Marks, ec='white', bins=sturges, zorder=2,
    label="students")
7
8 plt.xlabel('Marks')
9 plt.ylabel('Frequency')
10
11 plt.axvline(mean-std, linestyle='--', color='#F9AA33', label =
    "mean - std")
12 plt.axvline(mean, linestyle=':', label = "mean")
13 plt.axvline(mean+std, linestyle='--', color='#8EBA42', label =
    "mean + std")
14
15 plt.legend()
16 plt.grid(zorder=0, linestyle=':', alpha=0.6)
17
18 plt.show()

```

Output:



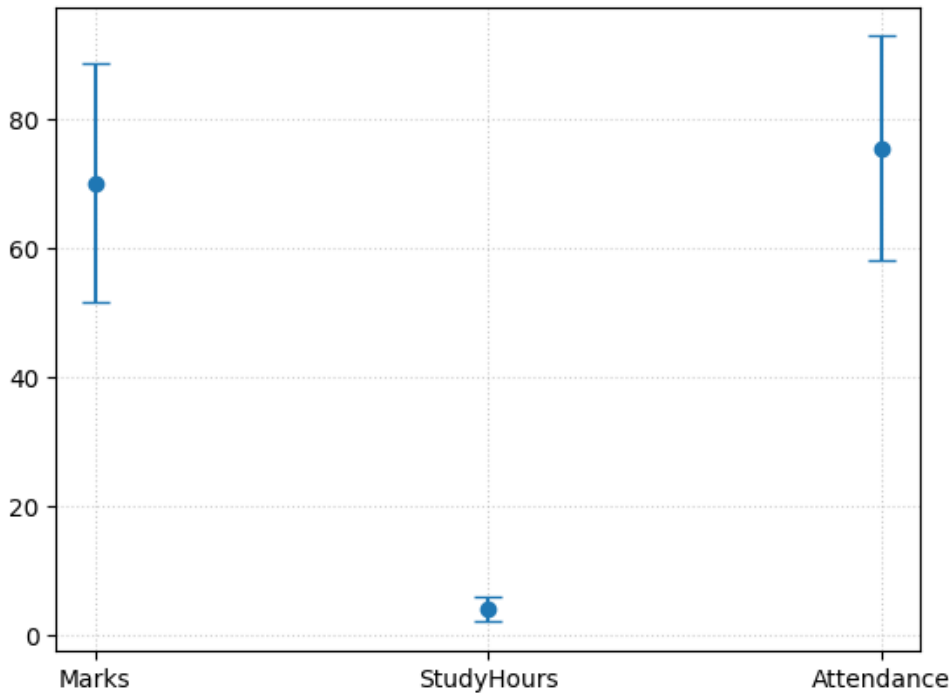
ii) Majority distribution in each attribute

```

1 mean = df_student[['Marks', 'StudyHours', 'Attendance']].mean()
2 std = df_student[['Marks', 'StudyHours', 'Attendance']].std()
3
4 plt.errorbar(mean.index, mean.values, std.values, fmt='o',
5               capsize=6, zorder = 2)
6
7 plt.grid(zorder=0, linestyle=':', alpha=0.6)
8 plt.show()

```

Output:



4.3. Observation

- The range of Marks is 60, StudyHours is 6 and Attendance is 57. Marks and Attendance have a wider spread compared to StudyHours.
- The standard deviation of Marks is 18.59, meaning most students scored within ± 18.59 of the mean, i.e. between 51.6 & 88.8. This can be seen in graphs above
- StudyHours has lowest variance and standard deviation, indicating that study hours is relatively consistent across students compared to thier marks or attendnace.
- Attendance has a varaince of 301.36 and standard deviation of 17.36, suggesting some inconsistency in how regular students atteneded class.
- High standard deviation of marks suggest that class is heterogeneous, i.e. performance varies from student to student.

Experiment 5

Calculate measure of variability such as range, variance and standard deviation for a dataset. Interpret what these measure indicate about the spread of the data.

5.1. Calculating measure of variability

i) Range

range = max - min

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 df_student = pd.read_csv("dataset.csv", delimiter=',',
6                           header='infer')
7
8 minval = df_student[['Marks', 'StudyHours', 'Attendance']].min()
9 maxval = df_student[['Marks', 'StudyHours', 'Attendance']].max()
10
11 rng = pd.DataFrame({
12     'Attribute': minval.index,
13     'Minimum': minval.values,
14     'Maximum': maxval.values,
15     'Range': (maxval - minval).values
16 })
17 print(rng)
```

Output:

	Attribute	Minimum	Maximum	Range
0	Marks	35	95	60
1	StudyHours	1	7	6
2	Attendance	42	99	57

ii) Variance

```
1 print(df_student[['Marks', 'StudyHours',
2                   'Attendance']].var().round(3))
```

Output:

Marks	345.752
StudyHours	3.995
Attendance	301.357

iii) Standard Deviation

```

1 df_student[['Marks', 'StudyHours', 'Attendance']].agg(['std',
    'mean']).T.round(3)

```

Output:

	std	mean
Marks	18.594	70.200
StudyHours	1.999	4.067
Attendance	17.360	75.567

5.2. Representation

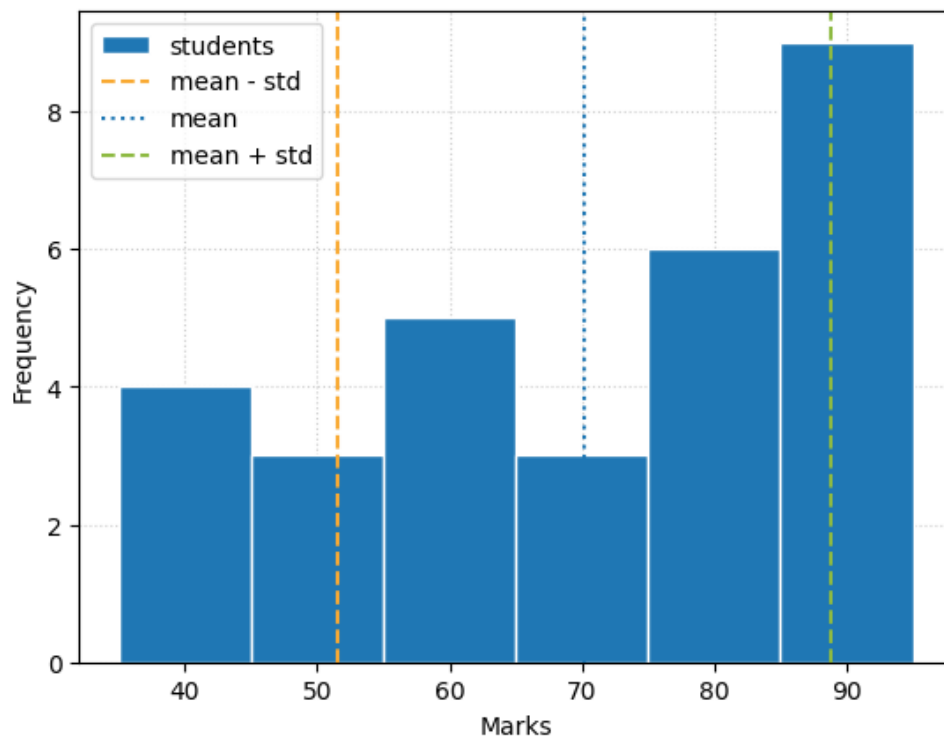
i) Spread of Marks

```

1 sturges = round(1+np.log2(df_student.Marks.count()))
2
3 mean = df_student.Marks.mean()
4 std = df_student.Marks.std()
5
6 plt.hist(df_student.Marks, ec='white', bins=sturges, zorder=2,
    label="students")
7
8 plt.xlabel('Marks')
9 plt.ylabel('Frequency')
10
11 plt.axvline(mean-std, linestyle='--', color='#F9AA33', label =
    "mean - std")
12 plt.axvline(mean, linestyle=':', label = "mean")
13 plt.axvline(mean+std, linestyle='--', color='#8EBA42', label =
    "mean + std")
14
15 plt.legend()
16 plt.grid(zorder=0, linestyle=':', alpha=0.6)
17
18 plt.show()

```

Output:



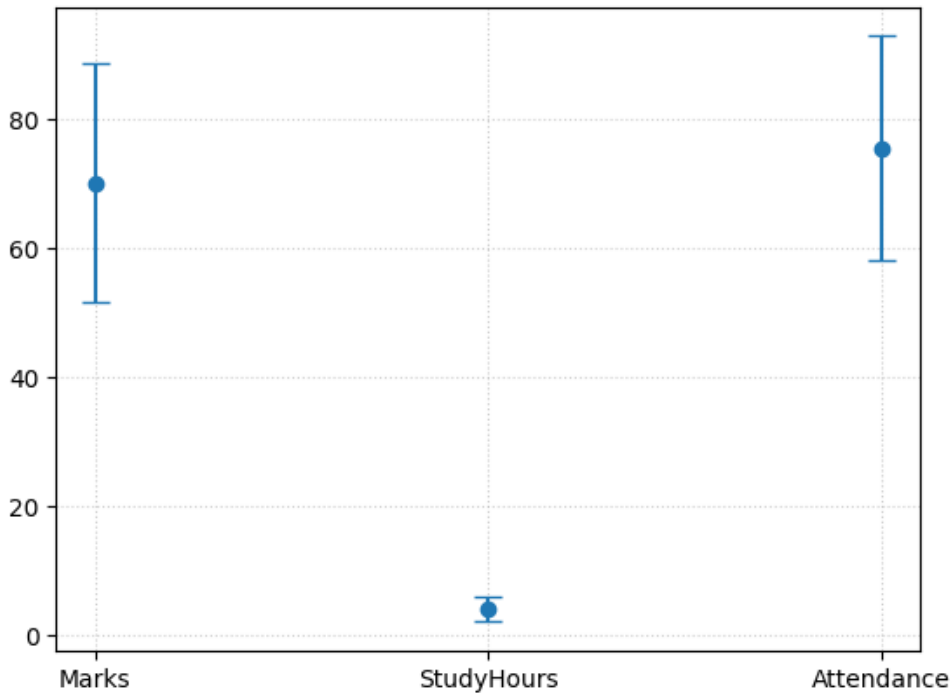
ii) Majority distribution in each attribute

```

1 mean = df_student[['Marks', 'StudyHours', 'Attendance']].mean()
2 std = df_student[['Marks', 'StudyHours', 'Attendance']].std()
3
4 plt.errorbar(mean.index, mean.values, std.values, fmt='o',
5             capsize=6, zorder = 2)
6
7 plt.grid(zorder=0, linestyle=':', alpha=0.6)
8 plt.show()

```

Output:



5.3. Observation

- i. The range of Marks is 60, StudyHours is 6 and Attendance is 57. Marks and Attendance have a wider spread compared to StudyHours.
- ii. The standard deviation of Marks is 18.59, meaning most students scored within ± 18.59 of the mean, i.e. between 51.6 & 88.8. This can be seen in graphs above
- iii. StudyHours has lowest variance and standard deviation, indicating that study hours is relatively consistent across students compared to thier marks or attendnace.
- iv. Attendance has a varaince of 301.36 and standard deviation of 17.36, suggesting some inconsistency in how regular students atteneded class.
- v. High standard deviation of marks suggest that class is heterogeneous, i.e. performance varies from student to student.